

To the Northwoods Police Department and the Special Investigator,

I want to extend my deepest gratitude for the tireless work you have put into Eleanor's case. While nothing can bring her back, I find comfort in knowing that the truth has finally come to light. For so long, there were more questions than answers, and that uncertainty weighed heavily on all who cared for her.

I now know that it was Margaret who was responsible, and while that knowledge is painful, it is also necessary. Eleanor deserved the truth, and so did all of us who loved her. I will carry the sorrow of that betrayal, but also the relief of no longer living in the shadows of doubt.

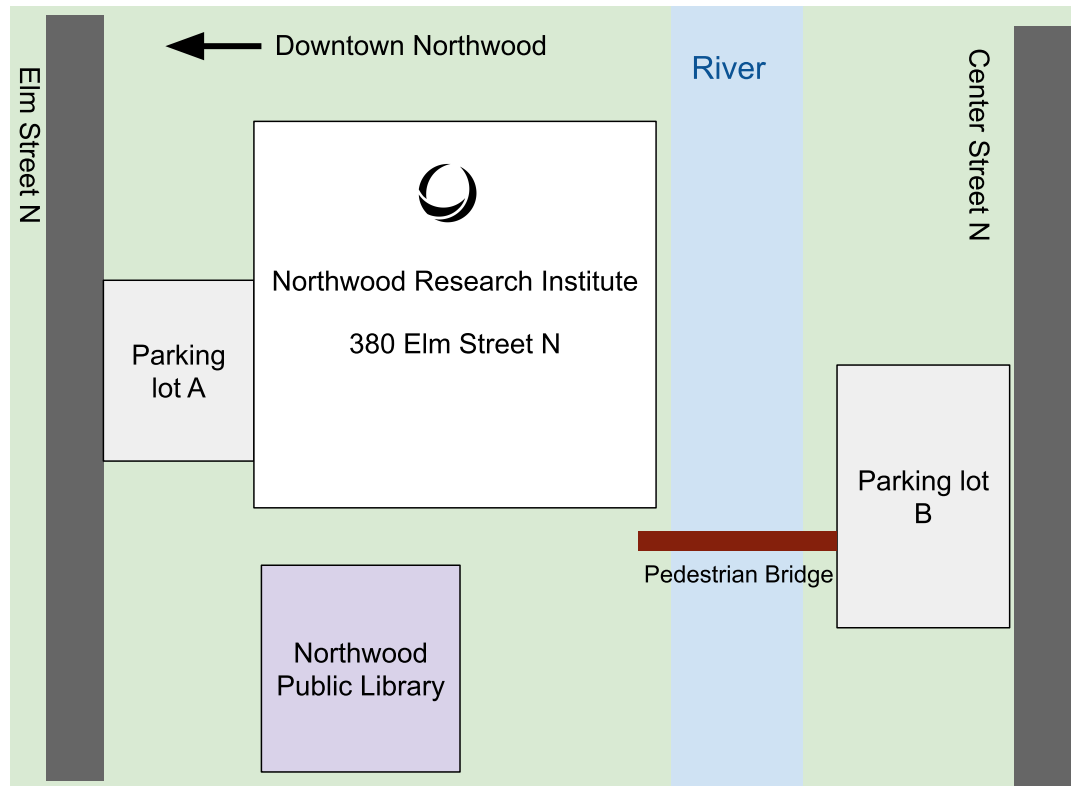
Your dedication, persistence, and willingness to look deeper when the easy answer was "accident" have made all the difference. Knowing how Eleanor's life truly ended brings me peace, even in sorrow. Justice, in its truest form, is not always about punishment but about truth — and now, at last, we have that truth.

Please know that your efforts will not be forgotten. Eleanor's memory is being honored not only by those who loved her, but by the students and colleagues who will carry her work forward. Because of you, her story is no longer defined by mystery, but by resolution.

With respect and gratitude,

A handwritten signature in cursive script that reads "Ann Calloway". The ink is black and the signature is fluid and elegant.

Ann Calloway



Scholarship Established in Memory of Dr. Eleanor Vance

By: Lydia Harper

NORTHWOOD — Northwoods Institute has announced the creation of a new scholarship honoring the late Dr. Eleanor Vance, a beloved scientist, mentor, and colleague whose contributions to biochemistry left a lasting mark on both research and education.

The scholarship, designed to support graduate students in biochemistry, is being made possible through a collaboration between three people who knew Eleanor best: Dr. Marcus Adler, fellow researcher; Catherine Rowe, former collaborator; and Ann Calloway, Eleanor's longtime partner.



Catherine Rowe and Marcus Adler worked closely with Northwoods Institute to finalize a new research partnership agreement that addresses many of Eleanor's priorities and concerns, ensuring her vision for responsible, impactful science will live on. A portion of the revenue generated from this partnership will be dedicated to funding the scholarship annually.

Meanwhile, Ann Calloway has taken on a pivotal role in outreach, helping with marketing and communications so that students across the region know about the opportunity. "Eleanor cared deeply about mentoring the next generation," Calloway said. "She believed science should be accessible, and this scholarship is one way we can honor her commitment."

Dr. Adler emphasized that the scholarship is not only about financial support but also about carrying forward Eleanor's values. "Eleanor demanded rigor, creativity, and compassion in her work. We hope this scholarship inspires students to pursue science with the same dedication she embodied."

Applications for the scholarship will open later this year, with details available on the Northwoods Institute website.

Through this initiative, Eleanor's colleagues and loved ones hope to ensure that her passion for biochemistry and her legacy of mentorship will continue to shape future scientists for years to come.



Northwood Police Department

Case #: 9649-2a

RE: Reopening of Case - Death of Dr. Eleanor Vance

On behalf of the Northwood Police Department, I want to personally thank you for your persistence and sharp eye in reviewing this case. Thanks to your efforts, we have identified critical details that were overlooked in the original conclusion.

As you pointed out, the compound Aetherium-monochloride (aka Chloroxyline) noted in the toxicology report—could not have reached lethal levels through inhalation. This fact was not public knowledge at the time, since Chloroxyline was a newly developed substance within the Northwood Laboratory and its toxicological profile had not yet been made widely available. Because of your contribution, we now recognize that Dr. Eleanor Vance's death was not an accident but a homicide.

We have already requested the security camera footage and badge access logs from the night of Dr. Vance's death. Retrieving and reviewing this material will take some time.

Next Steps in the Investigation:

- Rule out any suspects who could not have been at the lab at the time of the murder.

Your assistance has proven invaluable in ensuring justice for Dr. Vance. Please continue to share any insights or observations you uncover as the investigation moves forward.

With appreciation,

Jean Harris

Detective Harris
Northwood Police Department

Enhanced Catalytic Stability of Enzyme Complexes in Alternative Solvent Systems

Eleanor Vance, Ph.D.; Margaret Vance, Ph.D.

ABSTRACT

The activity of enzyme complexes is often constrained by solvent interactions, limiting catalytic efficiency under experimental conditions. We report that a class of oxidoreductase enzymes maintained higher stability and activity when transferred from traditional aqueous (dihydrogen monoxide) systems to a newly developed solvent medium, provisionally termed *Chloroxyline*. Early results suggest that this solvent reduces denaturation rates and prolongs active site availability.

INTRODUCTION

Enzyme stability remains a central challenge in biochemical applications. Conventional aqueous systems provide necessary solubility but frequently compromise long-term activity. Previous studies have noted partial improvements using buffered or ionic solutions, but denaturation remains problematic. In our work, we sought to evaluate an alternative solvent system that could preserve catalytic performance while maintaining compatibility with existing experimental methods.

METHODS

Purified oxidoreductase complexes were initially assayed in standard buffered dihydrogen monoxide. After baseline measurements, samples were transitioned to the experimental solvent system, *Chloroxyline*, developed within our laboratory. Enzyme kinetics, thermal stability, and structural retention were assessed using spectrophotometry and differential scanning calorimetry.

RESULTS

Relative to aqueous systems, *Chloroxyline* produced a measurable increase in catalytic half-life ($p < 0.05$) and reduced unfolding at elevated temperatures. Substrate turnover rates remained consistent, indicating compatibility without inhibition. Notably, structural integrity persisted even after extended exposure (>48h), suggesting improved solvent-protein interactions.

DISCUSSION

While dihydrogen monoxide has traditionally served as the universal medium, these findings indicate that *Chloroxyline* may represent a viable alternative for enzyme stabilization. The solvent appears to create a more favorable microenvironment, reducing denaturation events while supporting efficient catalysis. Future work will examine scalability and broader enzyme compatibility.

CONCLUSION

This preliminary study highlights the potential of *Chloroxyline* as a promising solvent in biochemical research. By enhancing enzyme resilience, it may enable longer and more reliable experimental runs. Continued investigation will determine its full applicability across systems.



Northwood Police Department

Autopsy Report (Form C-642)

Case No.
9649-2a

Date filed:
March 18, 2024

Victim Details

Name: *Eleanor Vance*

Age: *52*

Sex: *F*

Weight: *140 lbs*

Height: *5' 9"*

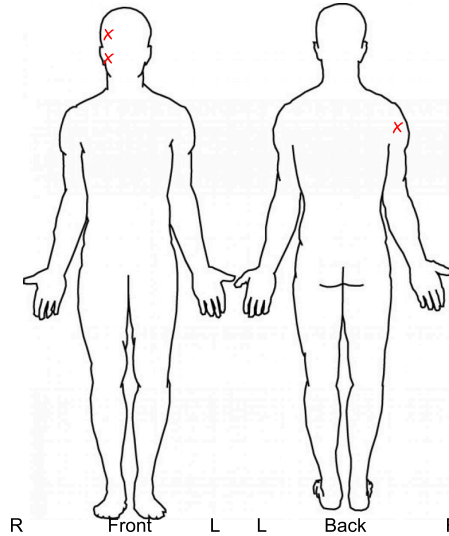
External Injuries

Fresh bruising found on the right side of face

Possible injection wound on right shoulder.

Internal Injuries

None



Cause of Death

Unknown at this time. Possible accidental poisoning by inhaling chemical vapors.

Identifying Information

No tattoos, scars or other identifying markings.

Body positively identified by Annette Calloway

Narrative

No signs of a struggle or other violence. The body was found next to broken glassware and an unknown chemical spilled on the floor. This chemical was not found on the victim's skin. A large bruise was forming on the right side of the face. A possible small injection wound on the right shoulder is consistent with a vaccine injection. All lab employees are required to get numerous vaccines as part of their employment.

A chemical smell was observed when the body was found. There are numerous chemicals in the lab that can, in certain instances, cause death if exposed to their vapors. A toxicology report has been requested.

Internal organs appear healthy and no other external or internal injuries were observed.

Pending the toxicology report, death appears to be accidental inhalation of toxic vapors after accidentally dropping a glass vial. Bruising on the face, most likely happened after loss of consciousness as the victim fell on the hard floor.

Based on degree of rigor mortis, and livor mortis, time of death is estimated between 10:30 pm and 11:15 pm on March 13, 2025

Coroner: *Dr. Pat Johnson MD*

Officer: *Det. Harris*

Signature: *Jean Harris*

Signature: *Dr. Pat Johnson MD*

Department: *NWPD*

Date: *03 / 18 / 2024*



Northwood Police Department

Case Notes

Case No.
9649-2a

Date filed:
April 19, 2024

CASE CLOSED

SUBJECT: Death of Dr. Eleanor Vance

Summary:

Following a full investigation into the death of Dr. Eleanor Vance, the Northwood Police Department has determined that the incident was accidental in nature.

Findings:

- Dr. Vance was discovered deceased in her laboratory on March 13, 2024.
- The coroner's examination found no signs of external trauma, assault, or struggle.
- Toxicology results indicated high levels of exposure to chemical fumes consistent with laboratory solvents.
- The crime scene showed broken glassware and a toppled stool, supporting the conclusion of an accidental spill.
- Fingerprints of lab personnel were found on common-use equipment and glassware, consistent with routine laboratory activity. No evidence of forced entry or tampering with security systems was found.
- Interviews with colleagues, family, and professional associates revealed motives for interpersonal conflict but produced no evidence of foul play. All suspects either had verifiable alibis or no demonstrated opportunity to access the laboratory at the time of death.

Conclusion:

The investigation concludes that Dr. Eleanor Vance was working alone in her laboratory and accidentally knocked over a vial of chemicals. She inhaled toxic fumes, became unconscious, fell, and subsequently died as a result of the accident.

Case Status: Closed - Accidental Death.

OFFICE

Detective Harris, Northwood PD

Jean Harris

From: d.keene@riverside.edu
To: e.vance@northwoodinstitute.org
Date: Mon, 4 Mar 2024 23:18:42 -0500
Subject: Enough

Eleanor —

I've tried to be collegial. I've tried to let the science speak for itself. But watching yet another review panel glom onto your name while my lab goes wanting is a constant, maddening reminder that the system rewards spectacle over sustained work.

You parade caution as if it were a virtue while the rest of us are expected to fight for every grant and every student. I'm tired, Eleanor. Tired of watching you collect accolades and funding while the rest of us rearrange scraps.

If you won't step aside or open the floor to a proper joint presentation — if you insist on hoarding the spotlight — then I'll make sure you stop getting it. I don't have time for niceties anymore. Either your reputation changes, or the field's attention will be redirected. I will do what I must to end your career one way or another.

This isn't personal — it's professional. You chose how you wanted to play this. Don't be surprised by the consequences.

—

Daniel V. Keene, Ph.D.
Head, Department of Biochemistry
Riverside University

From: Dr. Eleanor Vance evance@northwoodinstitute.edu
To: Catherine Rowe c.rowe@synbiocorp.com
Date: Thur, 29 Feb 2024 08:25:52 -0500
Subject: Contract Decision

Dear Catherine,

I wanted to take the time to personally explain my decision regarding the proposed contract. Up to this point, I have greatly valued our working relationship. SynBioCorp's support has provided my team with resources that made real progress possible, and I want to emphasize my gratitude for your professionalism and collaboration.

That said, after reviewing the details of the application outlined in the contract, I have serious ethical concerns. Specifically, the proposed use of enzyme-driven modifications falls into an area where safety testing, regulatory clarity, and potential misuse remain uncertain. While I understand the commercial potential, I cannot in good conscience move forward with an agreement that poses such risks.

I realize this decision has financial and professional implications, and it was not made lightly. Please know that this is not a reflection on you or SynBioCorp as a whole, but rather a necessary boundary given my responsibility as both a scientist and an educator.

I hope that despite this disagreement, we can maintain a positive relationship and continue exploring other avenues of partnership in the future. Your insight and commitment to science have always impressed me, and I remain optimistic that there are projects ahead where our values and goals will align.

With respect,
Eleanor Vance, PhD
Northwood Institute of Biochemistry



Northwood Research Institute

Material Safety Data Sheet (MSDS) – Simplified

Substance Name: Ammonium Hydroxide

Common Use: Cleaning agent, reagent

1. Identification

Clear, colorless liquid

2. Hazards

- **Corrosive/Irritant:** Burns skin, eyes
 - **Inhalation:** Strong irritation, coughing, difficulty breathing
 - **Ingestion:** Stomach pain, vomiting
-

3. First Aid

- **Skin Contact:** Wash with water
 - **Eye Contact:** Flush with water, seek medical care
 - **Inhalation:** Fresh air, medical help if severe
 - **Ingestion:** Rinse mouth, seek immediate medical help
-

4. Toxicity

Can enter the blood stream by both ingestion and inhalation. Blood levels rarely reach above 250 ng/ml by inhalation.

Toxic blood level 420 ng/ml



Northwood Research Institute

Material Safety Data Sheet (MSDS) – Simplified

Substance Name: Chloroxyline

Common Use: Reagent, Solvent

1. Identification

Clear, colorless liquid

2. Hazards

- **Inhalation:** Irritation, dizziness
 - **Ingestion:** Dizziness, nausea, intoxication
-

3. First Aid

- **Skin Contact:** Wash with water
 - **Eye Contact:** Flush with water, seek medical care
 - **Inhalation:** Fresh air, medical help if severe
 - **Ingestion:** Rinse mouth, seek immediate medical help
-

4. Toxicity

Can enter the blood stream by both ingestion and inhalation. Blood levels rarely reach above 1 ml/L by inhalation.

- **Inhalation:** Few symptoms
- **Ingestion:** May cause burns to stomach

Toxic blood level (8 ml/L)



Northwood Research Institute

Material Safety Data Sheet (MSDS) – Simplified

Substance Name: Dihydrogen Monoxide

Common Use: Universal solvent, cleaning, cooling

1. Identification

Clear, colorless liquid

2. Hazards

- Generally safe. Large spills may cause slips/falls
 - Contaminated samples may carry pathogens
-

3. First Aid

- No special measures required
-

4. Toxicity

Can enter the blood stream by both ingestion and inhalation. Considered safe for both ingestion and inhalation of vapor under normal use.

Toxic blood level N/A



Northwood Research Institute

Material Safety Data Sheet (MSDS) – Simplified

Substance Name: Ethanol

Common Use: Solvent, disinfectant, fuel additive

1. Identification

Clear, colorless liquid

2. Hazards

- **Flammable:** Vapors may ignite
 - **Ingestion:** Dizziness, nausea, intoxication
 - **Inhalation:** Irritation, dizziness
-

3. First Aid

- **Skin Contact:** Wash with soap and water
 - **Eye Contact:** Rinse with water
 - **Inhalation:** Move to fresh air
 - **Ingestion:** Seek medical attention if large amount consumed
-

4. Toxicity

Can enter the blood stream by both ingestion and inhalation. May enter the blood stream faster via inhalation of fumes. Requires large exposure (both inhalation and/or ingestion) before blood levels reach toxic levels.

Toxic blood level (200 mg/dL)



Northwood Research Institute

Material Safety Data Sheet (MSDS) – Simplified

Substance Name: Hydrochloric Acid

Common Use: Laboratory reagent, cleaning, chemical synthesis

1. Identification

Clear, colorless to slightly yellow liquid

2. Hazards

- **Corrosive:** Causes severe burns to skin, eyes, and mucous membranes
 - **Inhalation:** Irritation to respiratory system, coughing, choking
 - **Ingestion:** Severe internal damage, potentially fatal
-

3. First Aid

- **Skin Contact:** Rinse immediately with water for 15 minutes
 - **Eye Contact:** Rinse thoroughly with water; seek medical help immediately
 - **Inhalation:** Move to fresh air; seek medical attention if symptoms persist
 - **Ingestion:** Do NOT induce vomiting; seek immediate medical attention
-

4. Toxicity

Normally does not enter the blood stream via inhalation or ingestion

- **Inhalation:** May cause chemical burns to esophagus and lungs.
- **Ingestion:** May cause burns to esophagus and stomach.

Toxic blood level N/A



Northwood Police Department

Case #: 9649-2a

RE: Murder of Dr. Eleanor Vance

On behalf of the Northwood Police Department, I want to thank you once again for your continued assistance in the investigation into the death of Dr. Eleanor Vance. Your careful attention to detail has helped us make significant progress.

Through your efforts, we have been able to rule out both Daniel Keene and Catherine Rowe, as it has been confirmed that neither could have been present at the Institute at the time of Dr. Vance's death. This leaves us with three remaining persons of interest:

- Marcus Adler
- Renee Carter
- Margaret Vance

We have attached the most recent badge access logs and parking lot records for your review. Upon initial examination, nothing appears overtly unusual. However, given the complexities of this case, subtle details may be key.

At this stage, we are relying on your insight to piece together the available evidence and determine who was responsible for Eleanor's death — and how they committed the crime. Your unique perspective has already proven invaluable, and we believe you may be able to see connections that others have overlooked.

Your help could be the difference in bringing justice for Eleanor Vance.

With gratitude,

Jean Harris

Detective Harris
Northwood Police Department



Northwood Police Department

Case Notes

Case No.
9649-2a

SUBJECT: Homicide of Dr. Eleanor Vance

CASE CLOSED

Summary:

Following a full investigation into the death of Dr. Eleanor Vance, the Northwoods Police Department has determined that the incident was the result of a deliberate act carried out by Margaret Vance. The case is now closed.

Findings:

- Dr. Eleanor Vance was discovered deceased in her laboratory on March 13, 2024.
- Initial reports suggested an accidental chemical exposure. However, subsequent review by an outside investigator highlighted inconsistencies that warranted reopening the case.
- Evidence revealed that Margaret Vance took Dr. Vance's security badge from her personal belongings earlier that evening, providing unauthorized access to the laboratory.
- Once inside, Margaret Vance administered a dose of Chloroxyline via syringe. The substance, not widely known at the time, was staged to appear as though it had been spilled accidentally.
- Margaret Vance confessed to the act, stating her intention was to incapacitate, not kill, her sister. She admitted to staging the laboratory to conceal her involvement.
- Security logs, parking lot records, and witness statements were critical in reconstructing the sequence of events.

Conclusion:

The investigation concludes that Dr. Eleanor Vance's death was a homicide caused by the intentional administration of Chloroxyline by Margaret Vance. While her stated intention was not to cause death, her actions directly resulted in the fatality.

Case Status: Closed - Homicide (Confession obtained from Margaret Vance).

OFFICE

Detective Harris, Northwood PD

Jean Harris



Performance Review – Marcus Jonathan Adler

Department of Biochemistry, Northwood Research Institute

Marcus Adler has continued to demonstrate his capability as a brilliant scientist with strong potential for groundbreaking contributions in the field. Over the past year, he has authored two significant papers: “*Novel Pathways in Enzyme Stabilization Using Synthetic Polymers*” (*Journal of Molecular Biochemistry*, March 2022) and “*Cross-Linking Mechanisms in Rare Catalytic Proteins*” (*International Review of Chemical Sciences*, October 2022). Both works have received positive attention from peers and positioned Marcus as a promising voice in biochemical research.

However, concerns remain regarding his inattention to detail and tendency to rush processes. This has led to several lab accidents over the past year and, most notably, the retraction of his paper “*Accelerated Binding Kinetics of Aetherium-Derived Compounds*” (*Frontiers in Biochemical Research*, July 2023). The retraction was the result of faulty data collection and analysis practices, highlighting the need for greater care in his methodology.

It is the recommendation of this review that Marcus continue to pursue his research but under closer mentorship. Specifically, pairing him with Dr. Eleanor Vance is strongly encouraged. Dr. Vance’s methodical, detail-oriented approach could help Marcus refine his own processes and ensure that his creative drive is balanced with the rigor required for safe, reliable, and reproducible science.

Overall Assessment:

Marcus Adler has a bright future in the field and the intellectual capacity to make lasting contributions. With appropriate mentorship and greater attention to detail, he is well-positioned to grow into one of the Institute’s leading researchers.

Dr. Harold Whitmore
Head of Biochemistry Department
Northwood Research Institute